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9.2.2 The characteristic polynomial

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? Lemma 9.2.2.6

3

"What we will see is that we will instead create algorithms that converge to t...

Homework 9.2.2.1

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Let

$$A = \begin{pmatrix} \alpha_{00} & \alpha_{01} \\ \alpha_{10} & \alpha_{11} \end{pmatrix}$$

be a nonsingular matrix. Show that

$$\begin{pmatrix} \alpha_{00} & \alpha_{01} \\ \alpha_{10} & \alpha_{11} \end{pmatrix}^{-1} = \frac{1}{\alpha_{00}\alpha_{11} - \alpha_{10}\alpha_{01}} \begin{pmatrix} \alpha_{11} & -\alpha_{01} \\ -\alpha_{10} & \alpha_{00} \end{pmatrix}.$$

☒ done



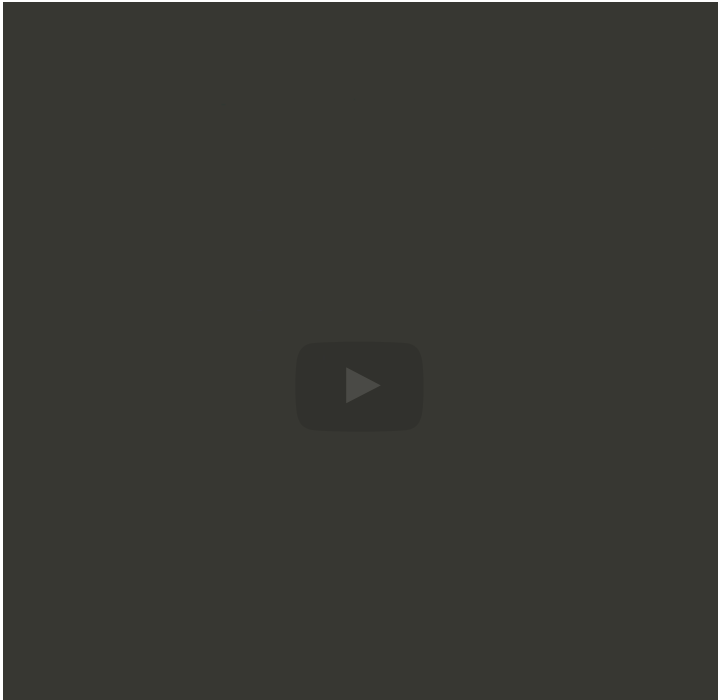
Solution:

For a complete solution see the [notes](#).

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Answers are displayed within the problem

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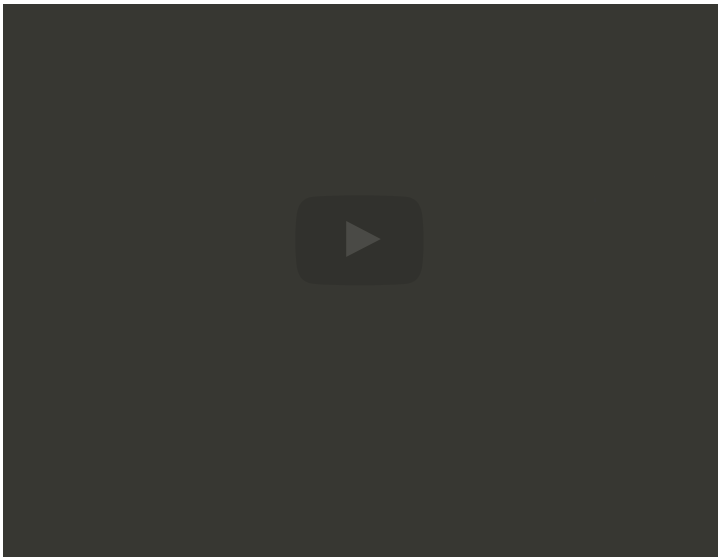
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we
use this to find the
eigenvalues of a two-
by-two matrix? Well,
we know that we
want to look at the
determinant of
lambda I minus A.
And that determinant
--
well, if we started
with a matrix
alpha_00,
alpha_01, alpha_10,
alpha_11, then this
matrix here is the
matrix lambda minus
that, minus
that , minus that, and
lambda minus that.
Alright. And then if
we apply the formula
for the determinant

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Hmm. Let's look at
the determinant a
little bit more. We
look at the
determinant of this 2



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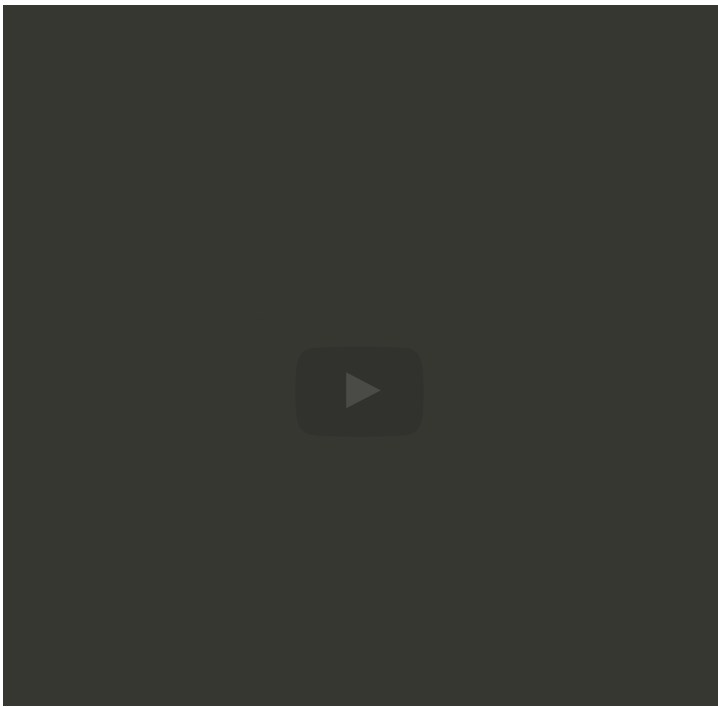
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by 2 matrix and we think of this as column a_0 and column a_1 .

These are vectors then the determinant turns out to be the area of the following parallelogram. If you take this and you take your vector a_0 and you root it at the origin. And you take the vector a_1 and you do the same thing

Video



λ can have a coefficient. So in other words any polynomial that has an,

you know, the coefficient associated with the first term equal to 1, you can construct a matrix that has that polynomial as its characteristic polynomial. And therefore, if we could come up with a formula for finding the

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